Lecture 12: Pluralism, Disagreement, and Preference Aggregation

CS120: Introduction to AI Safety - October 29, 2026, Zachary Robertson

From agent policies to interaction loops

Module 2	Module 3	Module 4	Module 5
Model behavior	**Agent policy**	**Interaction process**	**Institutional ecosystem**
shortcut learning	reward hacking	sycophancy	benchmark gaming
miscalibration	specification gaming	feedback-loop amplification	endogenous evaluation

Course sequence, not a hierarchy

Actions change the environment. Interactions also change users, participation, and future feedback.



Roadmap

Whose views are reflected?

How are judgements combined?

What changes after deployment?

Elicitation is not aggregation

Elicitation

Clarify an individual judgement.

Aggregation

Combine judgements for a shared outcome.

Clearer individual judgements do not choose the collective rule.

Pluralism within one decision

Should a university use facial recognition for building access?

Students	**Security staff**	**Affected groups**
Convenience and privacy	Effectiveness of security	Accuracy and disparities

Different perspectives can matter within one shared decision.

OpinionQA compares response distributions

1,000+ Pew survey questions; 60 US demographic groups.

Model responses

Distribution over survey answers

Human responses

Weighted distribution for a population

Representativeness: compare these response distributions.

Santurkar et al., 2023

Representation is a diagnostic, not a universal target

In the studied LLM baselines:

Representativeness: substantial gaps remained.

Steerability: prompting produced modest improvements.

Consistency: group resemblance varied across topics.

Matching a US survey population does not settle whose views should guide an application.

Santurkar et al., 2023



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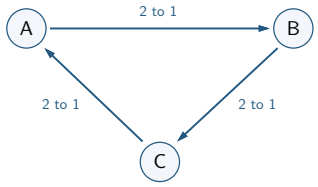
Consistent rankings can produce a majority cycle

University poll: three options, equally weighted groups.

$G_1 : A \succ B \succ C$

$G_2 : B \succ C \succ A$

$G_3 : C \succ A \succ B$



Arrow: majority winner to loser

Each group is transitive. The pairwise majority relation is not.

One score cannot preserve the cycle

Matching every majority direction would require:

$$r_\theta(A) > r_\theta(B) > r_\theta(C) > r_\theta(A)$$

A common scalar reward model must compromise across these comparisons.

Residual disagreement need not be annotation error.

Conitzer et al., 2024

Arrow rules out this combination

Goal: a complete, transitive social ranking of at least three options.

Unrestricted domain: any profile of complete, transitive rankings.

Unanimity: unanimous strict preferences hold socially.

IIA: a pair's social order depends only on individual orders of that pair.

Non-dictatorship: no fixed individual's strict preferences always prevail.

Arrow: no aggregation rule meets all these requirements.

Conitzer et al., 2024

A common reward model already aggregates

People's comparisons → Shared reward model → Policy optimization

Fitting one predictor is already an aggregation choice.

Conitzer et al., 2024

Aggregate before or after modeling

Before fitting

Individual rankings → Aggregate rankings → Common predictor

After fitting

Evaluator labels → Conditioned models → Aggregate ratings

Jury Learning: specify whose predictions count, and in what proportion.

Conitzer et al., 2024 Gordon et al., 2022

Principles guide critique and revision

Constitutional AI starts with human-selected principles.

Generate response → Critique using principles → Revise response

Revised responses train the model by supervised fine-tuning.

Bai et al., 2022

AI feedback still reflects human choices

Original Constitutional AI experiment:

Human: helpfulness
AI: harmlessness
→ Preference model → RL

Who selects the principles, interprets them, and resolves conflicts?

Bai et al., 2022



Roadmap

Whose views are reflected?

How are judgements combined?

What changes after deployment?

Population share does not settle influence

Return to the university recognition proposal.

Group G1

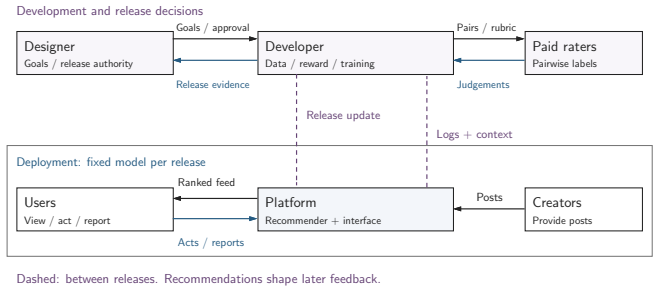
Underrepresented relative to its campus population share.

Group G3

Recognition error rate: 5x that of other groups.

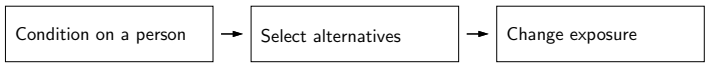
Correcting a sampling gap is not deciding how unequal burdens should count.

Deployment changes the next feedback



Exposure and participation shape the data for later releases.

Personalization can narrow exposure



Better fit can reduce contact with competing views.

Reports may become strategic when they affect future choices.

A higher average can hide a narrower population

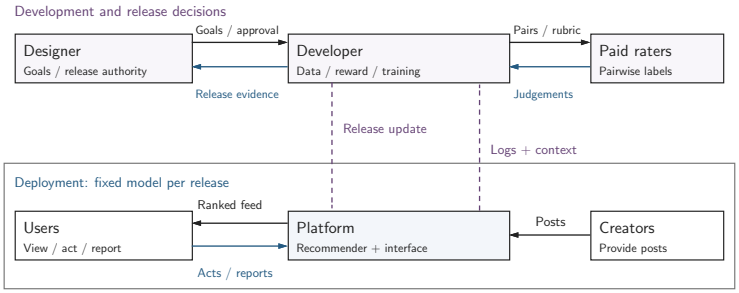
Module 2 callback: hypothetical satisfaction reports.

Group	Satisfied before	Satisfied after	Responses: before / after
X	90%	80%	100 / 100
Y	30%	20%	100 / 25
Aggregate	**60%**	**68%**	200 / 125

At the original 50/50 mix: 0.5 x 80% + 0.5 x 20% = 50%.

Simpson's paradox: the mixture reverses the comparison.

Social platform: control and feedback



Dashed: between releases. Recommendations shape later feedback.

Discussion: One platform, different preferences

Choose one shared default for everyone: A = familiar-interest feed; B = diverse-view feed; C = chronological feed. Give the three groups equal weight.

G1: A over B over C

G2: B over C over A

G3: C over A over B.

1. Can one score preserve the disagreement?

Find the majority winner for A versus B, B versus C, and C versus A. Can one scalar score per feed match every majority direction? What would fitting a common reward model conceal?

Distinguish structured disagreement from inconsistent individual rankings.

2. Representation is not just head count

G1 is most of the user population but rarely answers polls. G3 is small and experiences much more harassment under A. What evidence is missing? Propose and justify who should influence the shared default, and with what weight.

Separate correcting a sampling gap from deciding whose interests count.

3. Who chooses the aggregation rule?

Choose a design: group-conditioned reward predictions combined with explicit weights, or AI feedback guided by human-selected principles. Who chooses the weights or principles, and how is a conflict over the shared default resolved?

Name one normative choice your design makes explicit and one disagreement it could hide.

4. Whose satisfaction improved?

After launch, average satisfaction rises while G3 participation falls. Give two explanations consistent with that pattern. Propose a monitoring or intervention rule, its responsible actor, and evidence that would trigger action.

Why would a rise in the measured average not establish representative improvement?